Assignment - 03

Relational Algebra and SQL

1. Give the details of items having the price > 1000.
   Relational Algebra: $\sigma_{saleprice > 1000}(items)$

   Query:

```
---1. Give the details of items having the price > 1000.
SELECT *
FROM sales.items
WHERE saleprice > 1000;
```

   Output:

	code [PK] smallint	name character varying (20)	category integer	saleprice integer	stock integer	reorderlevel integer	averagepurchaseprice integer
1	1101	Printer	1	5000	100	20	4300
2	1105	Monitor	5	5500	100	10	4600

2. Give the details of items having the price > 1000 and belonging to category 5.
   Relational Algebra: $\sigma_{saleprice > 1000 \wedge category=5}(items)$

   Query:

```
---2. Give the details of items having the price > 1000 and belonging to category 5.
SELECT *
FROM sales.items
WHERE saleprice > 1000 AND category = 5;
```

   Output:

	code [PK] smallint	name character varying (20)	category integer	saleprice integer	stock integer	reorderlevel integer	averagepurchaseprice integer
1	1105	Monitor	5	5500	100	10	4600

3. List the Invoices (number, date) of a customer id = 'C05'.
   Relational Algebra: $\pi_{invno,invdate}(\sigma_{customerid='C05'}(invoice))$

   Query:

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```sql
---3. List the Invoices (number, date) of a customer id = 'C05'.
SELECT invno, invdate
FROM sales.invoice
WHERE customerid = 'C05';
```

Output:

Data Output | Messages | Notifications

	invno [PK] integer	invdate date
1	5	2011-08-23
2	1	2010-06-30

4. List all items (code, qty, rate) for invoice number = 1.
   Relational Algebra: $\pi_{itemcode,qty,rate}(\sigma_{invno=1}(invoicedetails))$

Query:

```sql
---4. List all items (code, qty, rate) for invoice number = 1.
SELECT itemcode AS code, qty, rate
FROM sales.invoicedetails
WHERE invno = 1;
```

Output:

Data Output | Messages | Notifications

	code smallint	qty integer	rate integer
1	1101	20	10000
2	1102	30	6000
3	1103	100	25000

5. List the Invoices (number, date) of the customer named Harsh.

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Relational Algebra: $\pi_{invno,invdate}(\sigma_{name='Harsh'}(customer \bowtie_{custid=customerid} invoice))$

Query:

```
---5. List the Invoices (number, date) of the customer named Harsh.
SELECT i.invno, i.invdate
FROM sales.invoice i
JOIN sales.customer c ON i.customerid = c.custid
WHERE c.name = 'Harsh';
```

Output:

Data Output | Messages | Notifications

	invno [PK] integer	invdate date
1	5	2011-08-23
2	1	2010-06-30

6. List all invoices (inv no, date, customer rid) that have item code = 1103

Relational Algebra: $\pi_{invno,invdate,customerid}(\sigma_{itemcode=1103}(invoice \bowtie invoicedetails))$

Query:

```
---6. List all invoices (inv no, date, customer rid) that have item code = 1103.
SELECT i.invno, i.invdate, i.customerid
FROM sales.invoice i
JOIN sales.invoicedetails id ON i.invno = id.invno
WHERE id.itemcode = 1103;
```

Output:

Data Output | Messages | Notifications

	invno [PK] integer	invdate date	customerid character (3)
1	6	2011-08-23	C01
2	3	2011-07-26	C03
3	2	2011-07-05	C04
4	1	2010-06-30	C05

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7. List all items (inv no, item code, qty) that customer 'C05' has bought.
   Relational Algebra: $\pi_{invno,itemcode,qty}(\sigma_{customerid='C05'}(invoice \bowtie invoicedetails))$

   Query:

```sql
---7. List all items (inv no, item code, qty) that customer 'C05' has bought.
SELECT id.invno, id.itemcode, id.qty
FROM sales.invoicedetails id
JOIN sales.invoice i ON id.invno = i.invno
WHERE i.customerid = 'C05';
```

   Output:

Data Output | Messages | Notifications

	invno [PK] integer	itemcode [PK] smallint	qty integer
1	1	1101	20
2	1	1102	30
3	1	1103	100
4	5	1102	70
5	5	1101	20

8. List all customers (customer id, name, city) that have bought item 1101.
   Relational Algebra:
   $\pi_{custid,name,city}(\sigma_{itemcode=1101}(customer \bowtie_{custid=customerid} invoice \bowtie invoicedetails))$

   Query:

```sql
---8. List all customers (customer id, name, city) that have bought item 1101.
SELECT DISTINCT c.custid, c.name, c.city
FROM sales.customer c
JOIN sales.invoice i ON c.custid = i.customerid
JOIN sales.invoicedetails id ON i.invno = id.invno
WHERE id.itemcode = 1101;
```

   Output:

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Data Output | Messages | Notifications

	custid [PK] character (3)	name character varying (20)	city character varying (20)
1	C05	Harsh	Jaipur

9. List all items (item code, item name) sold on 2011-08-23.

Relational Algebra:

$$\pi_{code,name}(\sigma_{invdate='2011-08-23'}(items \bowtie_{code=itemcode} invoicedetails \bowtie invoice))$$

Query:

```
---9. List all items (item code, item name) sold on 2011-08-23.
SELECT DISTINCT it.code, it.name
FROM sales.items it
JOIN sales.invoicedetails id ON it.code = id.itemcode
JOIN sales.invoice i ON id.invno = i.invno
WHERE i.invdate = '2011-08-23';
```

Output:

Data Output | Messages | Notifications

	code [PK] smallint	name character varying (20)
1	1101	Printer
2	1102	Pen Drive
3	1103	Key Board
4	1104	Mouse

10. List all items (item code, item name) sold in month '2011-07' from category=3.

Relational Algebra:

$$\pi_{code,name}(\sigma_{category=3 \wedge invdate \geq '2011-07-01' \wedge invdate < '2011-08-01'}(items \bowtie_{code=itemcode} invoicedetails \bowtie invoice))$$

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Query:

```
---10. List all items (item code, item name) sold in month '2011-07' from category=3.
SELECT DISTINCT it.code, it.name
FROM sales.items it
JOIN sales.invoicedetails id ON it.code = id.itemcode
JOIN sales.invoice i ON id.invno = i.invno
WHERE it.category = 3
  AND TO_CHAR(i.invdate, 'YYYY-MM') = '2011-07';
```

Output:

Data Output | Messages | Notifications

	code [PK] smallint	name character varying (20)
1	1103	Key Board

11. List customers (customer id, name, city) who are from Gujarat and have bought item items from category=3.

Relational Algebra:

$$\pi_{custid,name,city}(\sigma_{state='GJ' \wedge category=3}(customer \bowtie_{custid=customerid} invoice \bowtie invoicedetails \bowtie_{itemcode=code} items))$$

Query:

```
---11. List customers (customer id, name, city) who are from Gujarat and have bought item item
SELECT DISTINCT c.custid, c.name, c.city
FROM sales.customer c
JOIN sales.invoice i ON c.custid = i.customerid
JOIN sales.invoicedetails id ON i.invno = id.invno
JOIN sales.items it ON id.itemcode = it.code
WHERE c.state = 'GJ' AND it.category = 3;
```

Output:

Data Output | Messages | Notifications

	custid [PK] character (3)	name character varying (20)	city character varying (20)
1	C03	Tom	Diu

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12. List the items (code, name) of items purchased by a customer named 'Dev' on the date 2011-08-23.

Relational Algebra:

$$\pi_{code,name}(\sigma_{name='Dev' \wedge invdate='2011-08-23'}\ (customer \bowtie_{custid=customerid} invoice \bowtie invoicedetails \bowtie_{itemcode=code} items))$$

Query:

```
---12. List the items (code, name) of items purchased by a customer named 'Dev' on the date 201
SELECT DISTINCT it.code, it.name
FROM sales.items it
JOIN sales.invoicedetails id ON it.code = id.itemcode
JOIN sales.invoice i ON id.invno = i.invno
JOIN sales.customer c ON i.customerid = c.custid
WHERE c.name = 'Dev' AND i.invdate = '2011-08-23';
```

Output:

